

Paper Club · Sep 9

Empiricist

11 + 3 minutes

H_emp · 2×2 factorial · GPU budget

arXiv 2512.06951 · 1st Place BEHAVIOR 2025

Seat 1 · Kanta Saito

H_emp · Empiricist Hypothesis (P0)

**A large share of published q_score lift is
closed-loop recovery + System 2 context,
not only train-time architecture.**

Gripper OFF

drops partial success
on grasp-heavy tasks
(~2.2x order)

System 2 OFF

raises order / confusion
failures on ambiguous
tasks

Both OFF

worst aggregate
partial success
(super-additive)

Motion / Control Spine · Not classical planning

Learned action-chunk FM + rolling receding-horizon + stage hierarchy + rule recovery



Closed-loop overrides (outer loop)

System 2 · Stage Voting

- 5-15 stages / task
- vote window = 3 ($\geq 2/3$ forward)
- rollback rules for aliasing

NON-MARKOVIAN context

Gripper Recovery Heuristic

- force open if closed in stage
- never closed in demos
- paper: $\sim 2.2 \times q$ on 13×3 eps

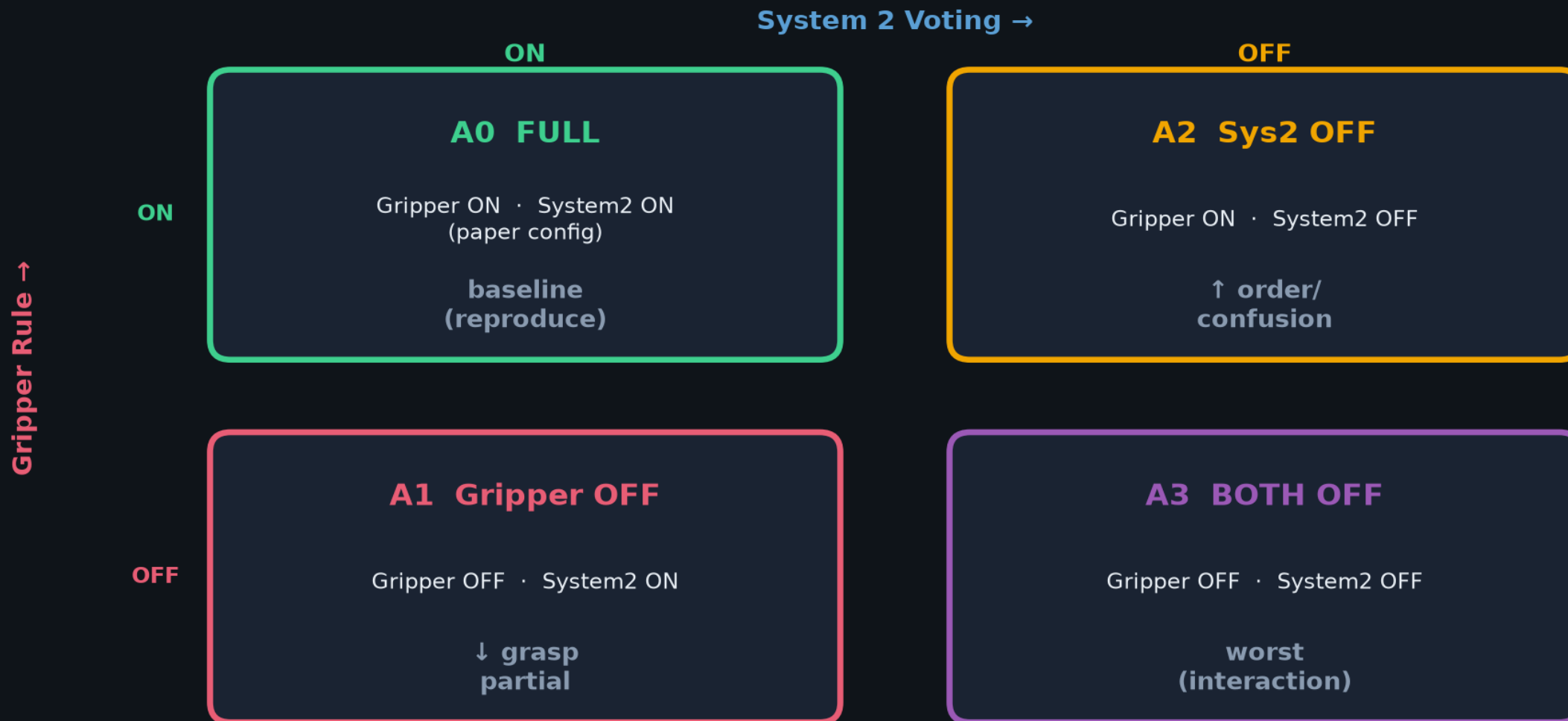
CLOSED-LOOP OVERRIDE

System 2 · Non-Markovian Stage Voting



E1 × E2 Factorial · THE Talk Figure

Inference-only on HF submission ckpts · same task subset × instances



HF Checkpoints · Suggested Eval Subset

IliaLarchenko/behavior_submission · task→ckpt map

Ckpt 1

20 tasks

2, 3, 5, 6, 10, 11, 13, 14,
15, 19, 23, 24, 25, 28,
29, 34, 42, 44, 47, 48

Ckpt 2

16 tasks

0, 1, 7, 8, 9, 12, 16, 17,
18, 20, 21, 22, 26, 30,
43, 45

Ckpt 3

13 tasks

4, 27, 31, 32, 33, 35, 36,
37, 38, 39, 41, 46, 49

Ckpt 4

1 task

40

Suggested subset (~8-12 tasks) × 3-5 instances × 4 conditions

Grasp-heavy / dexterity-dominated · Ambiguous stages (radio, microwave/popcorn)

Mix short vs long demo duration · Align with failure pie (Fig. 10)

Optional: compression OFF (E5) as negative control — should NOT dwarf E1/E2

Env / Run · Policy Server + Eval + Ablation Matrix

serve_blk.py

```
uv run scripts/serve_blk.py \  
  --task-checkpoint-mapping \  
  task_checkpoint_mapping.json \  
  policy:checkpoint \  
  --policy.config pi_behavior_blk_fast \  
  --policy.dir $CKPT_DIR
```

eval.py (websocket)

```
python BEHAVIOR-1K/omnigibson/\  
  learning/eval.py \  
  log_path=$LOG_PATH \  
  policy=websocket \  
  task.name=$TASK \  
  model.host=$HOST \  
  eval_instance_ids="[$IDS]"
```

run_ablation_matrix.sh · condition flags

gripper

ON / OFF

System2

ON / OFF

compression

1.0 / 1.3×

exec_horizon

10-26

GPU / RAM Budget · E1-E3 (inference-only)

No 8xH200 month · HF submission ckpts + OmniGibson eval subset

Smoke

1 task × 1 inst × 2 cond

2-6 GPU-h

Full P0 matrix

~10 tasks × 4 cond × 3-5 inst

~40-120 GPU-h

Policy VRAM

serve_b1k on 4090-class

~24 GB

Eval sys RAM

leaks; don't pack sims/node

≥48-64 GB

Success Criteria · Sep 9 Empiricist

PASS

Directional gripper drop on grasp-heavy
(order-compatible with $\sim 2.2\times$)

PASS

System 2 $\rightarrow$ failure-mode shift
($\uparrow$ order / confusion on ambiguous)

PASS

$A3 \leq \min(A1, A2)$ on aggregate
partial success

NOT a gate

Do NOT require global $q = 0.26$
Training ablations remain P2